

Enhancing City-Scale Road Extraction via Diffusion-Based Inpainting

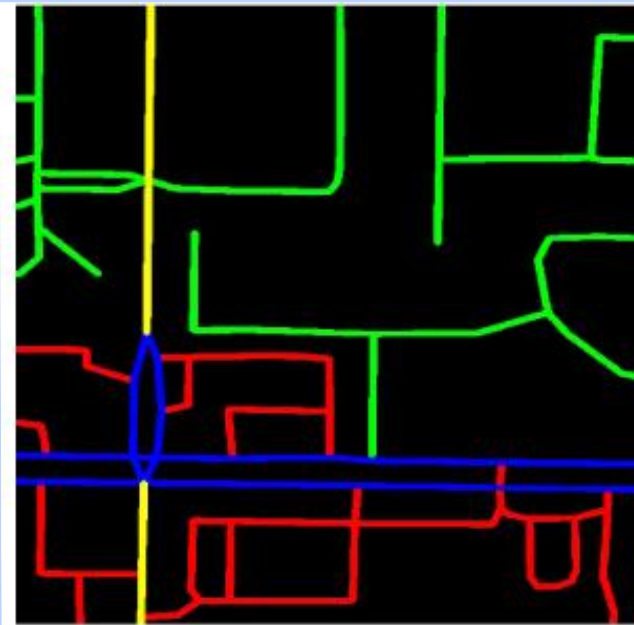
- Niels Versteeg, Yanick Palmers, Group 7



Universiteit
Leiden

Context, Cresi

- Extracts road networks from satellite imagery
- Estimates travel times and speed limits
- Resnet34 + U-Net



(a) Ground truth mask



(b) Predicted mask



(c) Ground truth network



(d) Predicted network

Problem Statement

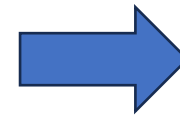
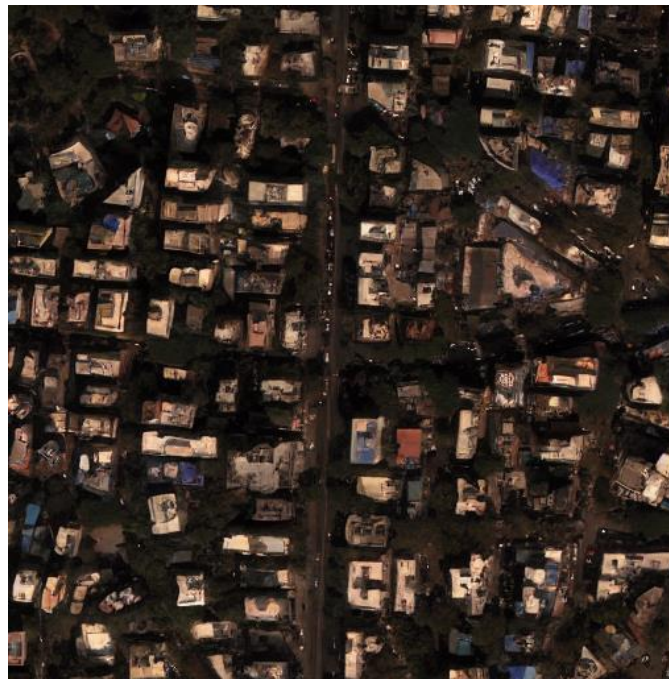
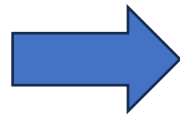
- **The Problem:** The "Perfect Image" Assumption
- CRESIv2 performs well on clear imagery but struggles with **occlusions:** clouds, dense tree canopies, and building shadows



Proposal: Pre-processing via Inpainting.

We try to add a **generative inpainting step** prior to the segmentation network.

Reconstruct missing road geometry and texture to "repair" occlusions.



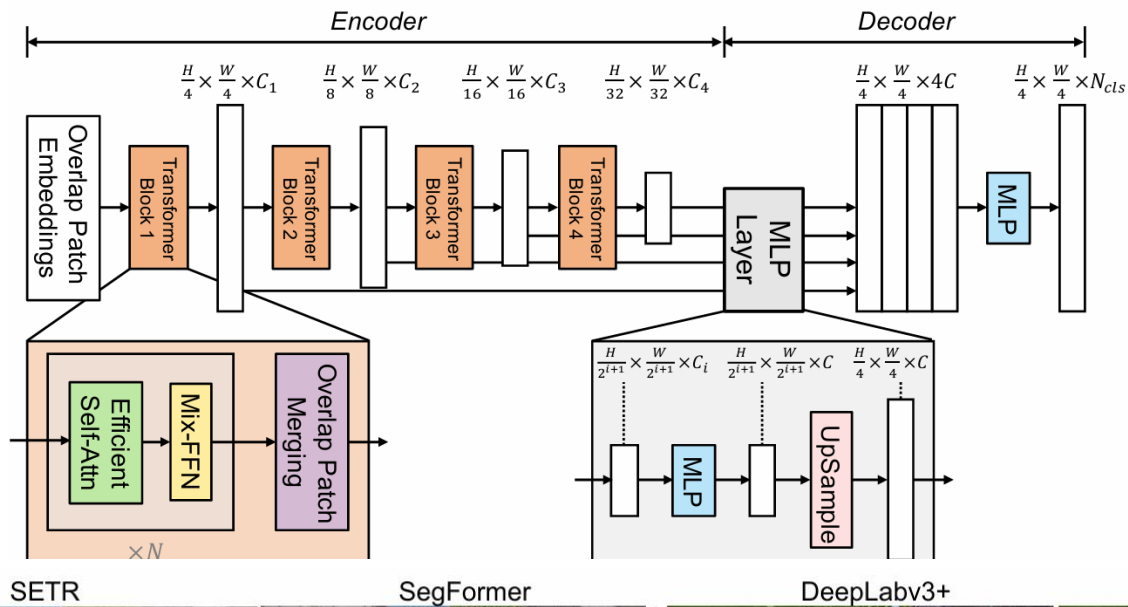
Methodology: The Pipeline

- Original:
 - **Input:** Raw Satellite Image (potentially occluded).
 - **Segmentation:** Feed image into CRESIv2.
 - **Graph Extraction:** Extract road speeds and vectors
- Our proposal
 - **Input:** Raw Satellite Image (potentially occluded).
 - **Pre-processing:**
 - Identify low-confidence/occluded regions (Mask Predictor).
 - Inpaint using Generative Diffusion Model.
 - **Segmentation:** Feed cleaned"image into original CRESIv2.
 - **Graph Extraction:** Extract road speeds and vectors

Methodology cloud masks

NVIDIA segformer, a pre-trained general segmentation model

Advantage: Able to generalize to other occlusions, like trees, shadows, etc.

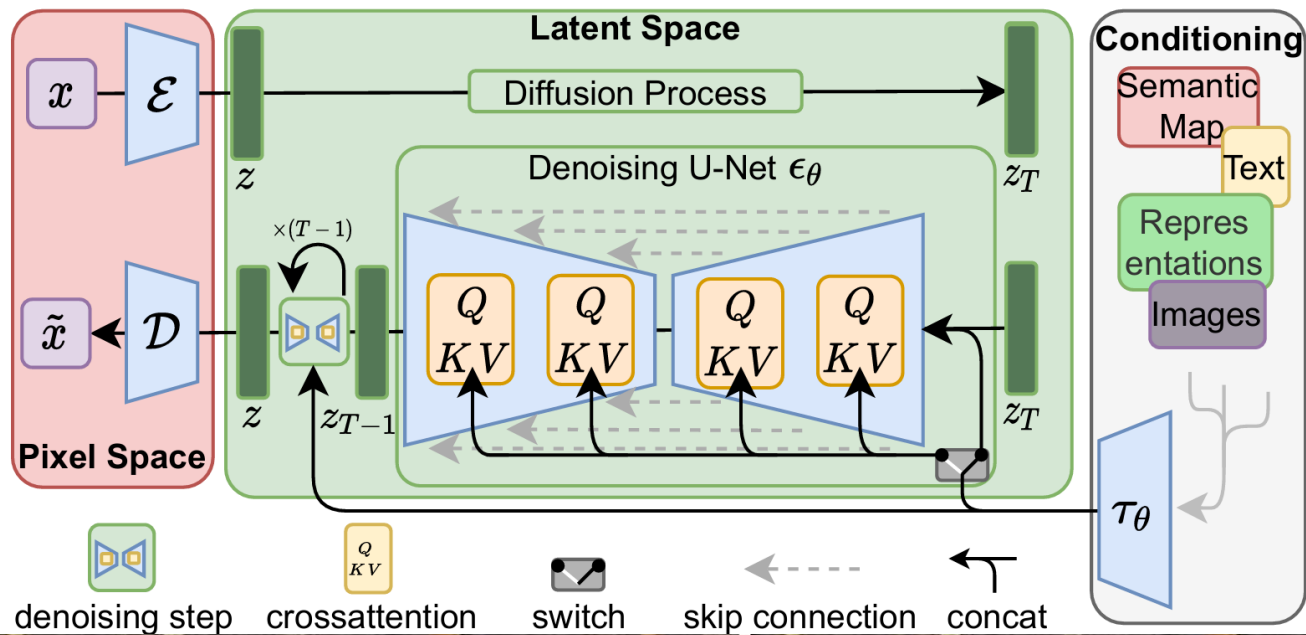


Original Satellite Image



Overlaid Cloud Mask





Methodology in-painting

- *Current Approach: DiffusionSat* (Diffusion Model)
- Trained for satellite imagery
- Add image, mask, and prompt



Current Progress

- Diffusion network
 - Implemented and running
 - Containerized setup
- Cloud mask model
 - Working
- Todo
 - Cloud mask output to in-paint model
 - Feed the output into CresiV2
 - Analyze performance

Metrics

Metrics (same as Cresiv2 Paper):

APLS (Average Path Length Similarity):

- Measures physical connectivity.

TOPO: Measures local graph topology.

Datasets:

- SpaceNet 3: loads of satellite images.
- Comparison: We will compare *Original CRESIv2* vs. *Inpainted + CRESIv2*.

Next steps

- Phase 1 (done)
 - Literature review, env setup, diffusion model working
- Phase 2 (doing now)
 - Add mask predictor
 - Test in-paining on SpaceNet examples
- Phase 3 (coming weeks)
 - Integrate in-paining results to Cresiv2
 - Run benchmarks
 - report

summary

- Road extraction is difficult with occluded images
- Remove occlusions from images by inpainting
- Expected outcome: higher accuracy on predicted road networks



Questions?